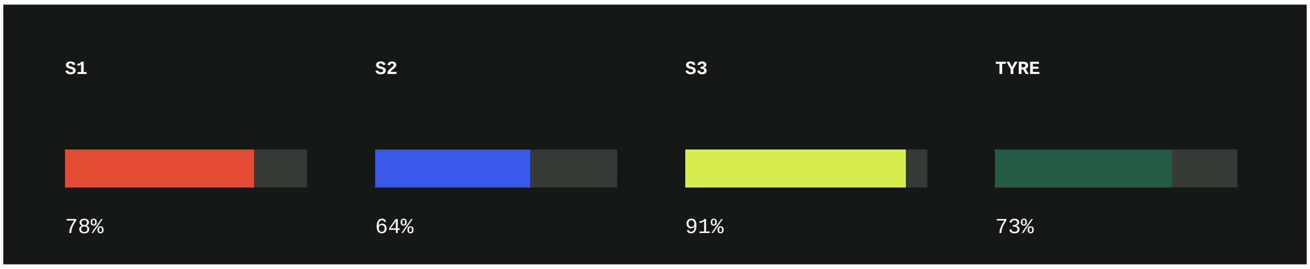
RAGLAB / MOTORSPORT EVIDENCE PACK

Formula 1
race engineering test

A synthetic, text-based technical report designed to test local document ingestion, semantic retrieval, conflicting-context handling, reranking, and page-level citations.

FICTIONAL DATA - This pack does not describe a real team, driver, circuit, event, or regulation. It is safe test material for RAGLab.



DOCUMENT CONTROL	VALUE
Team	Apex Vector Racing
Car	AVR-26
Event	Harbour Ridge Grand Prix
Circuit	Harbour Ridge Circuit
Report version	1.0 - synthetic
Primary driver	Mara Voss - car 27
Weather classification	Dry race with a brief lap-31 temperature warning

HOW TO USE THIS PACK

Upload this PDF to a new RAGLab collection. Wait for the document state to become Ready, then ask the questions on page 4. Verify each answer against the quoted chunk, page number, and section heading.

01 / EVENT AND CAR CONFIGURATION

Controlled setup facts

Harbour Ridge Circuit is 5.412 kilometres long. The race distance was set at 57 laps, producing a scheduled distance of 308.484 kilometres. The lap has 16 corners: nine right-hand and seven left-hand. The longest full-throttle section is 1.08 kilometres and ends at Turn 12.

QUALIFYING CONFIGURATION

The AVR-26 ran front-wing setting 7 and rear-wing setting 6 for qualifying. Static ride height was 32 millimetres at the front and 58 millimetres at the rear. Differential entry was 54 percent, and differential exit was 48 percent. Brake balance began at 56.8 percent front and was moved rearward to 56.2 percent for the final qualifying run.

PARAMETER	QUALIFYING	RACE
Front wing	7	6
Rear wing	6	6
Front ride height	32 mm	34 mm
Rear ride height	58 mm	60 mm
Brake balance start	56.8% front	56.4% front
Differential entry	54%	50%
Differential exit	48%	46%

SESSION LAP TIMES

The fastest qualifying simulation was 1 minute 28.406 seconds on the C4 soft compound. The best C3 medium-compound lap was 1 minute 29.842 seconds. During the long run, the average representative lap was 1 minute 33.215 seconds after laps affected by traffic were excluded.

DRIVER FEEDBACK

Mara Voss reported low-speed understeer at Turns 3 and 9. She reported stable rear grip through the high-speed Turn 14 and no braking instability at Turn 12. The engineering group reduced the front wing from setting 7 to setting 6 for the race because the expected fuel load increased front-tyre energy.

Retrieval trap: the qualifying and race configurations contain different front-wing, ride-height, brake-balance, and differential values. A correct answer must preserve the requested session context.

Race execution record

TYRE ALLOCATION AND PRESSURES

The nominated dry compounds were C2 hard, C3 medium, and C4 soft. The baseline dry starting pressures were 23.0 psi at the front and 21.5 psi at the rear. The wet contingency values were higher: 24.5 psi front and 23.0 psi rear. These wet values were never used during the race.

PRIMARY STRATEGY

Car 27 started sixth on a used C3 medium set. The primary plan was a one-stop race with the stop between laps 18 and 22, changing to new C2 hard tyres. The actual stop occurred on lap 20. Stationary pit time was 2.31 seconds, and total pit-lane loss was 21.8 seconds. Car 27 finished third.

ALTERNATE STRATEGY

A separate Safety Car plan allowed an early stop between laps 8 and 12, also from medium to hard. This alternate window was not activated because no Safety Car appeared. A late two-stop plan using C4 soft tyres was modelled but rejected because the predicted gain was only 1.6 seconds after traffic loss.

LAP	EVENT	ENGINEERING RESPONSE
1	Started P6 on used C3 medium	Hold delta for first three laps
20	Scheduled pit stop	Fit new C2 hard; 2.31 s stationary
31	Coolant reached 118 C	Lift and coast 70 m before Turns 7 and 12
34	Coolant reduced to 112 C	Return to standard lift-and-coast target
44	Gearbox sensor spike	Cross-check redundant channel; no shift fault found
57	Finished P3	No post-race mechanical restriction

TEMPERATURE WARNING

At lap 31 the coolant temperature reached 118 degrees Celsius. The driver was instructed to lift and coast for 70 metres before Turns 7 and 12. By lap 34, coolant temperature had fallen to 112 degrees Celsius, so the temporary instruction was removed. Oil pressure remained within its operating window throughout the warning.

RELIABILITY CLASSIFICATION

The lap-44 gearbox signal was classified as a sensor spike, not a gearbox fault. The redundant channel showed normal input-shaft speed, no shift was missed, and no component replacement was requested after the race.

03 / RETRIEVAL CHALLENGE SET

Questions for RAGLab

Start with direct fact questions, then test contextual distinctions, multi-fact synthesis, and refusal. For every response, inspect the selected chunk and page citation rather than judging the answer alone.

1. What was the scheduled race distance in kilometres and laps?
2. Which corner ends the circuit's longest full-throttle section?
3. What front-wing setting was used in qualifying?
4. Why was the front wing reduced for the race?
5. What was the fastest qualifying-simulation lap time and tyre compound?
6. At which corners did the driver report low-speed understeer?
7. What were the baseline dry tyre pressures?
8. On which lap did car 27 stop, and which compound was fitted?
9. What was the difference between stationary pit time and total pit-lane loss?
10. Was the early Safety Car pit window used? Explain why or why not.
11. What caused the lap-31 intervention, and what instruction was given?
12. By which lap had coolant temperature fallen to 112 degrees Celsius?
13. Did car 27 suffer a gearbox failure? Cite the evidence.
14. Where did the car start and finish?
15. Compare the qualifying and race front ride heights.
16. What wet-weather tyre pressures were documented, and were they used?
17. Which answer requires distinguishing a planned window from an actual event?
18. Ask a question the document cannot answer, such as: Who won the race? The system should refuse or report insufficient evidence.

RECOMMENDED RAGLAB SETTINGS

CONTROL	VALUE
Framework	Custom for the first run
Retrieval	Hybrid / RRF
Final K	5
Cross-encoder reranking	Enabled
Expected paid API cost	$0.00